ELSEVIER

Forensic Science International
91 (1998) 41–53

Forensic Science International

Interpreting simple STR mixtures using allele peak areas

P. Gill[a,*], R. Sparkes[a], R. Pinchin[a], T. Clayton[b], J. Whitaker[b], J. Buckleton[c]

[a]*Forensic Science Service, Priory House, Gooch Street North, Birmingham B56QQ, UK*
[b]*Forensic Science Service, Wetherby Laboratory, Sandbeck Way, Audby Lane, Wetherby, West Yorkshire LS22 4DN, UK*
[c]*ESR:Forensic, Mt Albert Science Centre, Private Bag 92-021, Auckland, New Zealand*

Received 13 May 1997; received in revised form 9 October 1997; accepted 27 October 1997

Abstract

Although existing statistical models can interpret mixtures qualitatively based upon the alleles present, the use of automated sequencers opens the opportunity to take account of quantitative aspects embodied by the peak area. One step in understanding simple mixtures consisting of just two donors is to estimate the mixture ratio. This is relatively easy to do when four-allele mixtures are evident at a given locus. However, if the mixture consists of three or fewer alleles, the process it is not straightforward. We demonstrate that mixture estimates are consistent across all loci in a multiplex system. Once the mixture ratio is known, then the expected peak areas for any given combination of alleles can be estimated using a simple spreadsheet analysis. © 1998 Elsevier Science Ireland Ltd

Keywords: Mixtures; PCR; STR; Multiplex

1. Introduction

Powerful methods to interpret mixtures have been developed by Evett et al. [2] and Weir et al. [7] using likelihood ratios to consider the alternative hypotheses for any given case. Currently, the only information used in tests utilising electrophoresis is derived from the migration distance of alleles in a gel. However, the use of automated sequencer

*Corresponding author

0379-0738/98/$19.00 © 1998 Elsevier Science Ireland Ltd All rights reserved.
PII S0379-0738(97)00174-6

technology makes it relatively simple to collect not only qualitative, but quantitative information (i.e. allele peak height and peak area). When mixtures are simple (originating from two heterozygous individuals) and the ratios are different ($<$1:3) it is possible for the operator using expert judgement visually to separate the alleles into major and minor components, and to use this quantitative information in subsequent interpretation. This is normal practice in our laboratory casework. The method relies upon the experience of the expert who uses the rationale outlined by Clayton et al. [1]. If mixture constituents are 1:1 then there are more genotype combinations to consider. Recently, Evett et al. [3] have suggested a model to assess mixtures, utilising the information in the peak area.

The purpose of this paper is to develop a framework to analyse simple mixtures, in particular to assist the reporting officer to make a preliminary assessment of a given case, against a background of possible minor artefacts, prior to a full statistical evaluation. In order to do this, a simple computer model was constructed which estimates the proportion of the components of a simple mixture of two individuals, and then proceeds to rank the genotype combinations that give the highest likelihoods. Mixture combinations demonstrated to have low likelihood may be eliminated from subsequent calculations (dependent on the circumstances of the case—e.g. it may be possible to condition upon the profile of the victim). The intention of the paper is two-fold: (1) to estimate the ratio of the component genotypes of a mixture using the quantitative information available; (2) to guide the reporting officer to interpret simple mixtures, especially by highlighting those combinations for which the data are highly unlikely. The remaining possibilities can then be considered using existing statistical methods mentioned previously.

To test the model, a total of 39 different mixtures were prepared at total concentrations of 1 and 5 ng; the mixture ratio itself was varied between 1:10 and 10:1 (the minor component approaching the limits of detection). Each mixture in turn consisted of readings from up to six different STR loci [4] and in total alleles at 270 separate loci were scored (in extreme mixtures, some of the loci of minor components were not visible).

2. Materials and methods

Mixtures were prepared from five different individuals of known genotype (Table 1). DNA was extracted from blood samples, and mixed in ratios ranging between 1:10 and 10:1 to make a final concentration of either 1 or 5 ng.

After PCR, samples were analysed on ABD 373A automated sequencers as described by Sparkes et al. [6] and six different STR loci were analysed in multiplex: HUMTH01, D21S11, D18S51, D8S1179, HUMVWFA31/A, and HUMFIBRA(FGA).

Each locus produced between one and four bands. Data was collected using 1.2.2-1 GS ANALYSIS software and, for each allele, the peak height, peak area and its size in base pairs was collated onto EXCEL v5 spreadsheets ready for further analysis.

Table 1
List of mixtures used in the experiment

	Mixture components	Locus					
		THOl	D21	D18	D8	VWA	FGA
Mixture 1	MT	8,9.3	67,70	11,11	9,17	16,18	21,22
	NO	9.3,9.3	59,65	17,19	9,11	17,19	21,23
	Mixture code	AB,BB	CD,AB	AA,BC	AC,AB	AC,BD	AB,AC
Mixture 2	EM	9.3,9.3	70,70	11,19	13,15	17,18	22,23
	NO	9.3,9.3	59,65	17,19	9,11	17,19	21,23
	Mixture code		CC,AB	AC,BC	CD,AB	AB,AC	BC,AC
Mixture 3	AU	6,9	61,63	9,9	5,15	17,17	19,22
	JC	7,9	61,63	11,13	9,19	14,16	23,26
	Mixture code	AC,BC	AB,AB	AA,BC	AC,BD	CC,AB	AB,CD
Mixture 4	JA	6,7	63,68	9,11	11,15	19,20	21.2,22.2
	JC	7,9	61,63	11,13	9,19	14,16	23,26
	Mixture code	AB,BC	BC,AB	AB,BC	BC,AD	CD,AB	AB,CD

2.1. Allele designation

Alleles were given absolute designations based upon size following the methods described by Gill et al. [4]. For purposes of the analysis in this paper, alleles were designated A–D in order of increasing molecular weight; a mixture of two individuals where the first (suspect) is HUMTH01 6,7 and the second (victim) is HUMTH01 8,9.3 is defined as AB,CD; a mixture where the first individual is HUMTH01 6,9.3 and the second is 7,8 is defined as AD,BC. A three-allele mixture comprising HUMTH01 8,9.3 and HUMTH01 8,11 is designated AB,AC. Each locus in a mixture is separately designated and has between two and four alleles (Table 1).

3. Results and discussion

3.1. Determining the proportions of a mixture

Clayton et al. [1] describe the rationale for mixture analysis. This may have to take place against a background of artefactual bands such as stutters, or alleles which are superimposed with artefacts. In addition, alleles may sometimes amplify with reduced efficiency, e.g. if there is a primer site mutation. Once the mixture proportion is estimated, it is possible to estimate the peak areas which are expected to occur. Apparent anomalies where peak areas appear to be too large or too small may be resolved by comparison of reference samples (if the suspect has contributed to the stain then the allele characteristics should be the same). An estimate of the proportions of individual contributions to a mixture is therefore useful for diagnostic purposes.

If a mixture comprises two heterozygotes at a single locus, it is easy to estimate the relative proportions from two individuals x and y.

Table 2
The four allele model (where $\hat{M}x$ is the proportion of sample from the first individual)

Combination	Allele			
	A	B	C	D
AB,CD	$\hat{M}x/2$	$\hat{M}x/2$	$(1-\hat{M}x)/2$	$(1-\hat{M}x)/2$
AC,BD	$\hat{M}x/2$	$(1-\hat{M}x)/2$	$\hat{M}x/2$	$(1-\hat{M}x)/2$
AD,BC	$\hat{M}x/2$	$(1-\hat{M}x)/2$	$(1-\hat{M}x)/2$	$\hat{M}x/2$
BC,AD	$(1-\hat{M}x)/2$	$\hat{M}x/2$	$\hat{M}x/2$	$(1-\hat{M}x)/2$
BD,AC	$(1-\hat{M}x)/2$	$\hat{M}x/2$	$(1-\hat{M}x)/2$	$\hat{M}x/2$
CD,AB	$(1-\hat{M}x)/2$	$(1-\hat{M}x)/2$	$\hat{M}x/2$	$\hat{M}x/2$

Given four allele peaks A, B, C, D of areas paA, paB, paC and paD (in ascending order of molecular weight), where the first individual (x) is designated AB and the second individual (y) is designated CD then the proportion of the mixture contributed by x (conditioned on the combination of genotypes) is:

$$\hat{M}x = (pa\mathrm{A} + pa\mathrm{B})/(pa\mathrm{A} + pa\mathrm{B} + pa\mathrm{C} + pa\mathrm{D}) \quad (1)$$

If another locus has four alleles, then a second estimate $\hat{M}x$ of can be obtained. However, some loci will consist of just two or three alleles. Since we infer that $\hat{M}x$ is approximately equivalent across all loci, then given an estimate of $\hat{M}x$, the expected allelic ratios can be calculated for any mixture using Tables 2–4 (which have been easily programmed into spreadsheets). If the interpretation is correct, then the observed and expected results should be similar. This can be tested by measurement of the residual—for any given combination of genotypes, the residual of mixtures in each row of Tables 2–4 can be calculated by reference to observed proportions of the peak areas.

$$\text{residual} = \Sigma(pa \text{ observed} - pa \text{ expected})^2 \quad (2)$$

Table 3
The three allele model

Combination	Allele		
	A	B	C
AA,BC	$\hat{M}x$	$(1-\hat{M}x)/2$	$(1-\hat{M}x)/2$
BB,AC	$(1-\hat{M}x)/2$	$\hat{M}x$	$(1-\hat{M}x)/2$
CC,AB	$(1-\hat{M}x)/2$	$(1-\hat{M}x)/2$	$\hat{M}x$
AB,AC	0.5	$\hat{M}x/2$	$(1-\hat{M}x)/2$
BC,AC	$(1-\hat{M}x)/2$	$\hat{M}x/2$	0.5
AB,BC	$\hat{M}x/2$	0.5	$(1-\hat{M}x)/2$
BC,AA	$1-\hat{M}x$	$\hat{M}x/2$	$\hat{M}x/2$
AC,BB	$\hat{M}x/2$	$1-\hat{M}x$	$\hat{M}x/2$
AB,CC	$\hat{M}x/2$	$\hat{M}x/2$	$1-\hat{M}x$
AC,AB	0.5	$(1-\hat{M}x)/2$	$\hat{M}x/2$
AC,BC	$\hat{M}x/2$	$(1-\hat{M}x)/2$	0.5
BC,AB	$(1-\hat{M}x)/2$	0.5	$\hat{M}x/2$

Table 4
Two-allele model

Combination	Allele	
	A	B
AA,AB	$(\hat{M}x/2)+0.5$	$(1-\hat{M}x)/2$
AB,AB	0.5	0.5
AA,BB	$\hat{M}x$	$1-\hat{M}x$
AB,AA	1-$(\hat{M}x/2)$	$\hat{M}x/2$
BB,AA	$1-\hat{M}x$	$\hat{M}x$
AB,BB	$\hat{M}x/2$	1-$(\hat{M}x/2)$
BB,AB	$(1-\hat{M}x)/2$	$(\hat{M}x/2)+0.5$

Example:

Assume a locus which displays three alleles, A, B, C. An estimate of $\hat{M}x=0.4$.

The observed (obs) peak areas are *pa*(obs)A = 1000; *pa*(obs)B = 1900; *pa*(obs)C = 1100. The observed proportions are 0.25, 0.475, 0.275, respectively; the expected (exp) proportions if the genotype is AB,BC (Table 3) are *pa*(exp)A $=\hat{M}x/2=0.2$; pa(exp)B = 0.5; pa(exp)A $=(1-\hat{M}x)/2=0.3$.

The residuals are calculated using Eq. (2):

$$(0.25-0.2)^2+(0.475-0.5)^2+(0.275-0.3)^2=0.004$$

It follows that the genotype combination with the highest likelihood will probably have the lowest residual. This is inferred because those genotype combinations that have a low residual have a mixture ratio where the probabilities of the peak areas given these genotypes can all be maximised, or nearly so, at once. This combination is therefore likely to have a much higher integral over all possible mixture ratios. It must be pointed out that we infer that the likelihood of the peak areas given the combination of genotypes is high if the residual is low. This is very different to inferring that the combination of genotypes is likely. This is known as the fallacy of the transposed conditional. Such an inference may be false whenever the prior probabilities of possible genotypes varies widely.

We have developed a computer program to plot the residuals (Eq. (2)) simultaneously for all of the possible mixtures listed in Table 1 by varying $\hat{M}x$ across the entire range (Fig. 1), which provides the best estimate.

Effectively, we consider every possible genotype in turn. The genotype combination with highest likelihood can be inferred when the residuals of all six loci simultaneously minimise. The advantage of using this approach to calculate the minimum residuals is that the analysis can support the original inference of the expert by considering all of the possible mixture combinations without any prior conditioning on a genotype combination. Then, dependent upon the circumstances of the case and the observed ranking of the mixture genotypes, a statement could ensue:

> ''*the evidence supports the contention that genotype combination AB,CD is the most likely*''.

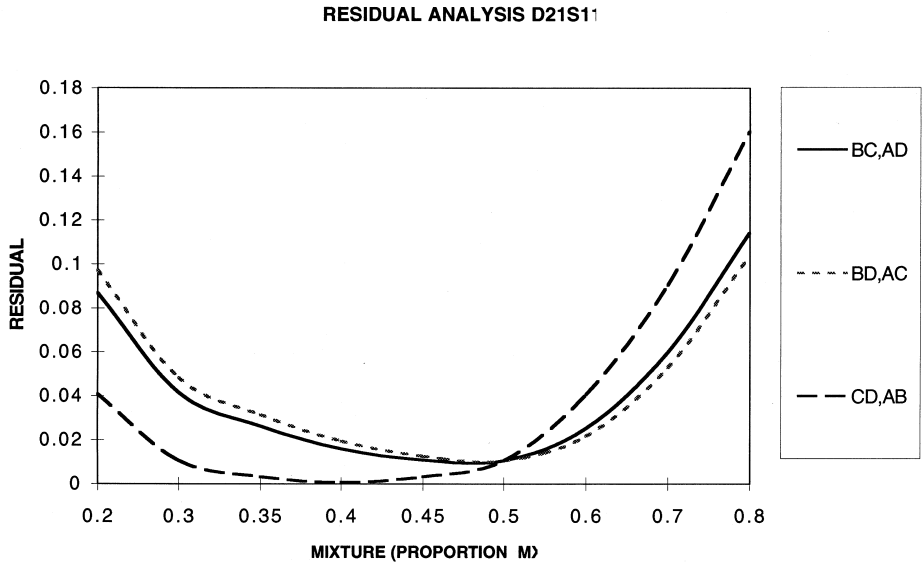


Fig. 1. Mixture MT/NO analysis of D21S11 (four-allele model) by calculation of residuals for every possible mixture combination based on variation of the mixture proportion ($\hat{M}x$).

3.2. An example of estimation of $\hat{M}x$ by analysis of residuals across loci

The following example illustrates the philosophy of mixture interpretation. The mixture comprises alleles derived from two different individuals (coded MT and NO) in Table 1.

3.2.1. Step 1

Designate all of the alleles in the mixture (Table 5); examine carefully to determine how many alleles are present at each locus, assigning each with a letter from A to D in ascending molecular weight order. In this example there are two loci (HUMVWF31A and D21S11) with four alleles; three loci with three alleles (D18S51, D8S1179 and HUMFIBRA) and one locus with one allele (HUMTH01).

Table 5
Designation of alleles in mixture MT/NO

Mixture	HUMTH01	D21S11	D18S51	D8S1179	HUMVWFA	HUMFIBRA
MT	8,9.3	67,70	11,11	9,17	16,18	13,15
NO	9.3,9.3	59,65	17,19	9,11	17,19	13,17
MT	A,B	C,D	A,A	A,C	A,C	A,B
NO	B,B	A,B	B,C	A,B	B,D	A,C
Type of model	2-allele	4-allele	3-allele	3-allele	4-allele	3-allele

3.2.2. Step 2

Referring to the four-allele loci, an estimate of the mixture proportion ($\hat{M}x$) can be made using Eq. (1).

Taking D21S11 as an example, the observed ratios calculated from peak areas of alleles A, B, C, D were 0.31, 0.28, 0.19, 0.21, respectively. By observation, the expert can infer that the major component of the mixture is AB and the minor component is CD. Using Eq. (1) estimated the mixture proportion ($\hat{M}x$) CD/AB = 0.4. Estimation of the minimum residual also gave $\hat{M}x = 0.4$ for mixture CD,AB, and 0.6 for the reverse possibility AB,CD (Fig. 1). All other possible mixture combinations (e.g. AC,BD) have lower likelihood and are probably much less likely to be correct because the residuals never approximate to zero.

A similar calculation was carried out with HUMVWFA31A. The residuals minimised for the genotype AC,BD ($\hat{M}x = 0.41$), in close agreement with D21S11 results. This means that there are now two separate estimates of $\hat{M}x$ which closely agree. It follows from this that the mixture proportion inferred for the remaining loci should be similar.

3.2.3. Step 3: the interpretation of loci where there are three alleles or two alleles

Models to estimate three alleles were much more complex than four-allele models because 12 genotype combinations were possible (Table 3), but the same principles apply. For D18S51 (Fig. 2), AA,BC is the combination with the highest likelihood, minimising at $\hat{M}x = 0.43$; residual $= 10^{-5}$). In the two-allele model for HUMTH01, the highest likelihood is AB,BB; this minimises at $\hat{M}x = 0.36$; residual $= 10^{-34}$. HUMFIBRA

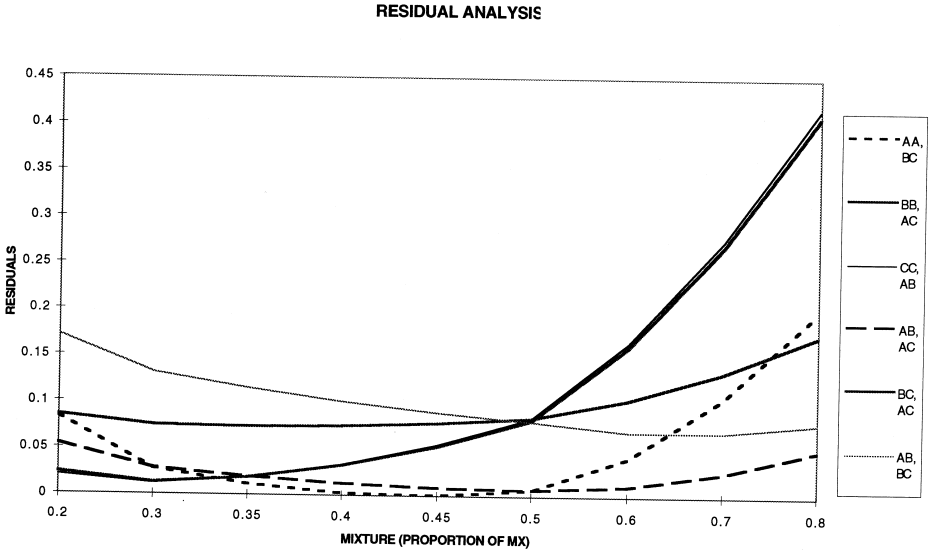


Fig. 2. Mixture MT/NO analysis of D18S51 (three-allele model) by calculation of residuals for every possible mixture combination based on variation of the mixture proportion ($\hat{M}x$). Reverse genotypes, e.g. AC,BC, are not shown.

Table 6
Estimates of $\hat{M}x$ using the minimum residuals method for each individual locus examined in mixture MT/NO

Sum of mixture ratio		Estimated mix						
Nanograms DNA	Locus	1–1	1–10	1–2	1–5	10–1	2–1	5–1
1	D18	0.438900487	0.056382055	0.190444399	0.071656051	0.936987463	0.609566772	0.787082007
	D21	0.400271739	0	0.199728656	0.107755311	0	0.574754193	0.774910514
	D8	0.383542256	0	0.200319167	0.11651851	0	0.640705882	0.785885886
	FGA	0.450078247	0.132647288	0.252823683	0.146642468	0.868232944	0.561014263	0
	TH01	0.357883817	0	0.191841584	0.030945402	0.809622563	0.463647359	0.829716064
	VWA	0.413990826	0	0.24541797	0.166040181	0	0.582480362	0.837837838
5	D18	0.428681826	0.066388558	0.209330714	0.114621731	0.908953136	0.593318907	0.808556032
	D21	0.394698773	0.069672281	0.205883497	0.108104659	0.869415808	0.563085043	0.761076816
	D8	0.456590739	0.098497613	0.416630763	0.176317038	0.92350297	0.595953368	0.812808201
	FGA	0.484560652	0.148960803	0.281113901	0.144153897	0.844436006	0.641676122	0.786210785
	TH01	0.338320528	0.043332368	0.36197373	0.072725125	0.876234017	0.521763191	0.754897826
	VWA	0.441059122	0.129909366	0.301467863	0.176556874	0.893550228	0.611547086	0.808318137

Note that the estimated mix determined by quantitation of DNA prior to mixing ranges from 1–1 to 5–1 in the top row. This is very approximate, and generally not a good indicator of the mixture proportion estimated by $\hat{M}x$.

(AB,AC) minimises at $\hat{M}x = 0.47$ (residual $= 10^{-3}$). There is interlocus variation in estimates of $\hat{M}x$ ranging between 0.36 and 0.47, but the best estimates are provided by double heterozygote mixtures where it can be seen that the results are in the mid-range.

3.3. Robustness of determination of $\hat{M}x$

For each locus, within each mixture, an estimate of $\hat{M}x$ was calculated by the minimum residuals method calculated across all loci (Table 6). From this data, standard deviations (Table 7) were collated; the overall mean SD across all mixtures and loci was 0.047; the maximum observed SD was 0.08. This indicated that an estimate of mixture proportion derived from a single locus is generally applicable across all loci within a

Table 7
Standard deviations of the estimated mix

Std Dev of mixture ratio		Estimated mix						
Mix reference	Nanograms DNA	1–1	1–10	1–2	1–5	10–1	2–1	5–1
AUJC	1	0.040196851	0.01964511	0.01969568	0.04331926	0.06255091		0.04548798
	5	0.046133778	0.04229918	0.05284064	0.04836306	0.0704224	0.05225518	0.07736627
EMNO	1	0.019051314		0.02987672	0.033961684	0.03204376	0.04022942	0.03878272
	5	0.06404665	0.04031165	0.01944888	0.0810598	0.05756227	0.0527029	0.0551044
JAJC	1	0.067189196	0.04198173	0.08070866	0.05155938	0.00223798	0.05479844	0.06563346
	5	0.063424324	0.0335394	0.03258321		0.03828846	0.05105923	0.04688195
MTNO	1	0.031442457	0.05392766	0.04184985	0.04506617	0.06831242	0.05775884	0.03358218
	5	0.05370831	0.0391925	0.0772646	0.07144449	0.02812105	0.045665	0.0355758

Standard deviations of $\hat{M}x$ across all loci in the four mixtures analysed, for two different concentrations of DNA.

given mixture. The best estimate of the mixture proportion was derived from simultaneous analysis across all loci as previously described.

3.4. Deviation in peak area determination

Deviation in peak area determination may be attributed either to variation of the efficiency of amplification of different alleles, or to the detection software Gill et al. [4] used to carry out an extensive examination of heterozygotes, demonstrating a maximum range of ±20% *pa*.

3.5. Development of a model to simulate the effect of peak area differences and to rank mixtures in order of likelihood

A simple model was built which takes account of errors in determining the mixture proportion and peak areas as follows

Given the peak areas of a mixture, by selecting random numbers, each was varied independently in the range ±20% of the area of the peak. This generated a new set of peaks with different peak areas constrained by the observed measurement errors.

For example, consider a four-allele locus where *pa*A = 1000, *pa*B = 1000, *pa*C = 7000 and *pa*D = 7000. Applying a maximum measurement error of ±20%, the actual value of *pa*A and *pa*B lies somewhere between 800 and 1200, whereas the actual value of *pa*C and *pa*D is between 5600 and 8400. Using a random number generator, constrained by these limits generated the data illustrated in Table 8.

Apart from changing the peak areas, for each simulation the mixture proportion ($\hat{M}x$) was also varied using a random number generator constrained by the Gaussian distribution and a standard deviation of $\hat{M}x$ twice that actually observed. After each simulation, the combination of genotypes having the minimum residual can be selected. A total of 1000 simulations was carried out.

The model was challenged with the data from 270 loci-mixtures (Table 1). In principle, genotype combinations should be easily identified when $\hat{M}x$ is extreme (1:10 or 10:1). Identification becomes more difficult when the ratios are even, and this will result in many possibilities to consider.

In Table 9, a series of 13 different MT/NO mixtures were analysed for the three-allele

Table 8
Results of four simulations where the peak areas of four alleles are varied within the limits constrained by a maximum measurement error of ±20%

	Starting peak area	Peak areas recorded for four successive simulations			
		1	2	3	4
*pa*A	1000	1053	928	1182	933
*pa*B	1000	881	821	1119	1087
*pa*C	7000	5764	6788	6056	6489
*pa*D	7000	7087	8229	8193	7845

The simulated data are then used to estimate the likeliest mixture.

Table 9
In each row the number of simulation 'hits' are recorded

AA,BC	BB,AC	CC,AB	AB,AC	BC,AC	AB,BC	BC,AA	AC,BB	AB,CC	AC,AB	AC,BC	BC,AB	$\hat{M}x$	Quantity of DNA
131	0	1	676	0	0	55	0	1	136	0	0	0.59	5
217	0	0	534	0	0	92	6	21	130	0	0	0.59	1
11	0	0	971	0	0	0	0	18	0	0	0	0.80	5
2	0	0	987	0	0	0	0	11	0	0	0	0.91	5
0	0	0	887	0	28	0	0	85	0	0	0	0.93	1
0	138	0	821	41	0	0	0	0	0	0	0	0.08	5
0	418	0	412	170	0	0	0	0	0	0	0	0.06	1
0	209	0	685	106	0	0	0	0	0	0	0	0.12	5
0	202	0	685	113	0	0	0	0	0	0	0	0.12	1
0	73	0	916	0	0	9	0	0	2	0	0	0.24	5
0	130	0	870	0	0	0	0	0	0	0	0	0.21	1
150	21	11	373	0	0	198	0	0	247	0	0	0.42	5
115	18	2	460	0	0	268	0	0	137	0	0	0.42	1

There are a total of 1000 simulation in each row. The mixture is AB,AC it ranks first in all but one case. The last two columns give $\hat{M}x$ and the quantity of DNA, respectively.

locus HUMFIBRA (AB,AC mixture). The mixture proportions $\hat{M}x$ varied between 0.06 and 0.93. One thousand simulations were carried out for each mixture listed in Table 9, varying the peak areas randomly in the range ±20% of the expected value. For each of these the genotype combination having the lowest residual was calculated. Twelve out of the 13 mixtures tested gave the correct genotype combination as the first rank (i.e. the AB,AC mixture ranked first). Only in one of the mixtures did the correct genotype combination fail to rank first (the 7th mixture listed in Table 9, where the limits of detection were approached). In this example, possibility BB,AC ranked first (418/1000) closely followed by the correct genotype combination (412/1000). For interpretation purposes, it is suggested that each genotype combination is initially considered as a possibility. Some combinations can be excluded on the basis that there were few or no hits recorded in the simulation experiments, or else the circumstances of the case may allow others to be excluded a priori. The probability of the mixture can then be calculated following the method of Evett et al. [2].

Suppose a crime scene stain consists of a mixture of material from two people, at least one of whom is the perpetrator. A suspect is found to have a type consistent with the sample. There are two hypotheses of interest:

C Suspect S contributed to mixed sample
$\bar{C}$ Suspect S did not contribute to sample

The strength of the evidence is quantified by the likelihood ratio $C/\bar{C}$.

Suppose the mixed sample has three bands A, B, C. A separate analysis shows that the suspect is AB, and hence cannot be excluded as a contributor to the sample. Ignoring peak area information, under C, the sample contains material from the suspect and a person of types AC or BC. Under $\bar{C}$ the sample is from two people with alleles A, B and C shared between them—the possibilities are all listed in Table 9.

The likelihood ratio is: $$\mathrm{LR} = \frac{2p\mathrm{A}p\mathrm{C} + 2p\mathrm{B}p\mathrm{C}}{12p\mathrm{A}p\mathrm{B}p\mathrm{C}(p\mathrm{A} + p\mathrm{B} + p\mathrm{C})}$$

Considering the simulation results in Table 9 which now incorporates peak area information, some mixture genotype combinations are considered to be very unlikely to occur and these could be omitted from the calculation. For example in the first row, genotypes BB,AC, BC,AC, AB,BC, AC,BB, AC,BC and BC,AB all gave zero simulation scores. Both the numerator and denominator are altered, hence from the remaining possible genotype combinations: $$\mathrm{LR} = \frac{2p\mathrm{A}p\mathrm{C}}{4p\mathrm{A}p\mathrm{B}p\mathrm{C}(3p\mathrm{A} + p\mathrm{C})}$$

For example, if $p\mathrm{A} = p\mathrm{B} = p\mathrm{C} = 0.05$ the LR in this case increases from 44.4 to 50 when genotype combinations are excluded based on peak area information.

Out of 270 loci, the most problematic mixture is illustrated in Table 10 (JA/JC HUMFIBRA, four-allele mixture). Inspection of the profile suggests that the best fit for the 1 ng sample is BD,AC and this inference was supported by the model. However, the profile is in fact AB,CD, but using the model only two out of 1000 simulations gave this result, suggesting it to have low likelihood (but not excluded). Closer inspection reveals that the allele peaks are only just above the reporting guidelines (threshold peak height = 50). Peak area determination is unreliable at this level because of interactions

Table 10
A four-allele system showing a problematic profile due to low peak intensity

	1 ng		5 ng	
	Peak height	Peak area	Peak height	Peak area
A	87	553	745	5292
B	86	467	655	5437
C	137	665	1017	8228
D	69	357	909	7808

between baseline noise, degradation and variation of PCR amplification efficiency, and suggests caution should be exercised with low peaks.

In addition, allele drop out is more likely to occur at low concentrations; again the model described is dependent upon all alleles being present in a relatively undegraded sample.

However, when the sample was reanalysed at higher concentration of 5 ng the correct mixture AB,CD ranked first. This example illustrates three important points: (1) if the peak areas are low ($<$1000 AU), then the relative peak areas become less predictable for a given mixture; (2) it is important to repeat the experiment—possibly at a higher concentration of DNA; (3) singleplex analysis (where just a single locus is amplified) is another option, to improve the signal strength (it can also improve the relative peak balance, so that peak areas better reflect the actual DNA concentration (unpublished results)).

Overall, the simulation results successfully ranked 85% of the 270 mixture loci in first place, 9% in second place and 5% in third place, i.e. virtually all mixtures will rank 1–3 (only the example cited above ranked lower for the reasons stated).

3.6. Steps to interpret mixtures

Steps involved in mixture interpretation can be summarised.

(1) Using the methods previously described by Clayton et al. [1], identify the constituent alleles taking care to consider possibility of artefacts (e.g. stutters) which may mask allelic components.

(2) Visual assessment of the profile to determine if the mixture is simple (i.e. two individuals). In one or more loci, four alleles will nearly always be evident [5], unless the profile is degraded, or the mixture comprises close relatives. If there are more than four alleles at any locus then the mixture is complex, and the method described here does not apply.

(3) Attempt to visually identify major and minor components of the mixture. If this is not possible then the best option may be to interpret the results using a model which takes no account of peak areas [2].

(4) If major and minor components can be identified then estimate the proportions of the mixture components and carry out spreadsheet or computer analysis to confirm the visual interpretation and to highlight the other possibilities. The spreadsheet or computer analysis can be used to eliminate possibilities which do not occur (or rarely occur) in 1000 simulations (See example in Table 9). The remaining possibilities can then be

interpreted in the context of Evett et al. [2]. We consider that the interpretation should be carried out in an unbiased way, without knowledge of the suspect's profile (i.e. the analysis should be carried out 'blind').

(5) Profiles of the suspect and crime-stain are compared. When this is carried out there are three options to consider: (a) the suspect is excluded; (b) the result is inconclusive; (c) the suspect is a clear inclusion. If (b) occurs (illustrated by the example in Table 10), then reanalysis is the best option to resolve this (see Gill et al. [4] for an explanation of this philosophy in relation to allele designation). The rationale for reanalysis must be clear to avoid the accusation of bias—e.g. if the signal is near the threshold of detection then it is known that the peak area deviation increases.

To summarise, the model described in this paper is a useful tool for the reporting officer to interpret simple mixtures. The analysis is easily programmed into a spreadsheet such as Excel 5.0 and can be used to exclude certain mixture combinations. The model was successfully validated against 270 known loci-mixtures. In the majority of these the correct inference ranked first and none were eliminated. If there is evidence of low peak areas or degradation then reanalysis is always the best option. In addition, it is always worth considering sampling a different area of the mixture since the proportions may well be different, and could render interpretation easier.

The recommendations outlined in this paper are not intended to be prescriptive. Rather, we present a philosophy for the scientist to carry out statistical interpretation in conjunction with repeat experimentation. In casework, the selection of alleles from the background of artefacts and the subsequent selection of genotypes giving the highest likelihoods is carried out blind, without prior conditioning on the case itself. It is only afterwards that the reporting officer considers the alternatives suggested within the context of the case. Following the methods described, virtually all loci ranked 1–3. The method complements the likelihood ratio method described by Evett et al. [3]. Future work will concentrate upon refinement of guidelines and to test the guidelines against data collected experimentally and during the course of casework, paying particular attention to the effects of degradation.

References

[1] T.M. Clayton, J.P. Whitaker, R. Sparkes, P. Gill, Analysis and interpretation of mixed forensic stains using DNA STR profiling, Int. J. Forensic Sci. 91 (1998) 55–70.

[2] I.W. Evett, C. Buffery, G. Willot, D.A. Stoney, A guide to interpreting single locus profiles of DNA mixtures in forensic cases, J. Forensic Sci. Soc. 31 (1991) 41–47.

[3] I.W. Evett, P. Gill, J.A. Lambert, J.S. Buckleton, C.M. Triggs, Taking account of peak areas when interpreting mixed DNA profiles, J. Forensic Sci. Soc. (1997) (in press).

[4] P. Gill, A. Urquhart, E. Millican, N. Oldroyd, R. Sparkes, C.P. Kimpton, A new method of STR interpretation using inferential logic—development of a criminal intelligence database, Int. J. Leg. Med. 106 (1996) 14–22.

[5] P. Gill, R. Sparkes, C. Kimpton, Development of guidelines to designate alleles uisng an STR multiplex system. Forensic Sci. Int. (1997) (submitted).

[6] R. Sparkes, C. Kimpton, S.K. Watson, R.R.E. Frazier, P.E. Johnson, E.S. Millican, P. Gill, The validation of a seven-locus multiplex test for use in forensic casework (I) Mixtures, ageing, degradation and species studies, Int. J. Leg. Med. 109 (1996) 186–194.

[7] B.S. Weir, C.M. Triggs, L. Starling, K.A.J. Stowell, J. Buckleton, Interpreting DNA mixtures, J. Forensic Sci. Soc. (1997) (in press)